

Frequently Asked Questions (FAQs) – RainSafeRoute

General & Working Questions

- **1. How does RainSafeRoute differ from mainstream navigation apps like Google Maps or Apple Maps?** Standard navigation apps prioritize time and distance, often routing unsuspecting drivers through waterlogged streets. RainSafeRoute specifically factors in live weather data, historical flood hotspots, and community reports to calculate a flood-risk score, prioritizing your *safety* over raw speed.
- **2. Where does RainSafeRoute get its live weather and rainfall data?** The platform integrates with reliable weather services like the Open-Meteo API to deliver real-time data on rainfall volume, wind speed, and humidity.
- **3. Can RainSafeRoute predict waterlogging before it happens?** Yes, by combining live meteorological data with historical monsoon trends and known urban drainage bottlenecks (such as Delhi's identified waterlogging hotspots), the system evaluates risk profiles for different paths.
- **4. What happens if I am traveling through an area with no internet connection?** Currently, RainSafeRoute requires internet connectivity to fetch live routing and weather updates, but offline emergency routing and fallback features are slated for future development.
- **5. Are the routes optimized for different types of vehicles?** Yes, the platform includes specialized travel modes tailored for cars, bikes, heavy trucks, school commutes, and emergency vehicles like fire trucks and police cars.
- **6. How accurate are the flood risk percentages given for each route?** The flood risk percentage is calculated using a multi-layered approach—aggregating live weather conditions, historical flood logs, and crowd-sourced community reports to yield a robust safety score out of 100.
- **7. Can municipal corporations or city authorities use this data?** Yes, RainSafeRoute features an analytics and admin dashboard designed to give insights into city-wide flood hotspots, user-generated incident reports, and daily route safety statistics.
- **8. Is RainSafeRoute available as a mobile application?** Currently, it functions as a responsive web-based platform built with React and Vite, with dedicated native Android and iOS mobile applications planned for future releases.
- **9. How many route options does the app provide when I search for a destination?** The system typically evaluates and generates up to 5 alternative route options, ranking them based on distance, ETA, and flood risk.
- **10. What technologies power the backend and mapping systems of RainSafeRoute?** The platform is built using a modern tech stack featuring React.js on the frontend, Node.js and Express.js on the backend, Supabase (PostgreSQL) for database management, and OpenStreetMap/OpenRouteService APIs for mapping and routing.

Privacy & Community Reporting Questions

- **11. Do I need to create an account or sign up to use the navigation features?** Basic source-to-destination route searching and safety score evaluations can be accessed

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smoothly, while community reporting features tie into user sessions to maintain data integrity.

- **12. When I report a waterlogged road, is my personal identity or phone number exposed to other users?** No, RainSafeRoute respects user privacy. Community reports display only the incident details—such as water depth (e.g., knee-level) and time stamps—without exposing personal contact information.
- **13. Are photos uploaded with community reports checked for private data (like license plates or faces)?** Optional photo attachments are meant strictly for verifying waterlogging severity. Users are encouraged to capture only street conditions, and the platform focuses analytics on safety metrics rather than personal surveillance.
- **14. Is my live GPS location tracked continuously by RainSafeRoute?** Your location data is used strictly in real-time to process your source and destination search parameters and provide accurate routing instructions; it is not permanently tracked or monetized.
- **15. How does the platform prevent fake or malicious waterlogging reports from users?** The system employs timestamp validation and data aggregation logic. Future updates aim to integrate municipal sensor verification to cross-check crowd-sourced hazards.
- **16. Where is my navigation search history and user data stored?** All application data is securely managed and stored using a scalable Supabase (PostgreSQL) database infrastructure with robust security protocols.
- **17. Do third-party advertisers get access to my travel routes or destination history?** No. RainSafeRoute is built as an impact-driven public safety platform under the AWS Summer Builder Cohort, and your travel patterns remain private.
- **18. Can I delete my community reports or interaction history from the system?** Yes, time-stamped incident reports and user contributions are managed dynamically through the platform's database framework.
- **19. Does the app track my speed or driving behavior while navigating?** No, RainSafeRoute focuses entirely on environmental hazards, route elevation, and flood risk analysis rather than monitoring personal driving speeds or behavior.
- **20. Who can I contact if I notice a major privacy or security concern regarding the platform?** You can explore the detailed system architecture, documentation, or reach out to the developers (Ria Saraswat and Vanshika Tyagi from Team Cipher, IGDTUW) through the project's official documentation links.